

1. How does Power BI handle large datasets in the Online Service, and what is the role of Premium Capacity in this?

How power bi handles large datasets depends on the storage mode. If it's import mode, then power BI handles it with its own engine VertiPaq and also premium provides limit extension up to 400 GB. If it's directquery mode then the source stays in the system, be it sql server, snowflake, data warehouse, and sends queries back to that source. Live Connection is much like DirectQuery mode, but it connects to Analysis Services or Power BI Datasets

2. What are the differences between Import mode, DirectQuery, and Live Connection in Power BI Service?

Import Mode: Data is copied from the data source into the server, so takes up memory. Works best for small to medium-sized datasets

DirectQuery Mode: Data stays in the source (SQL Server, Azure, Snowflake) and Power BI sends queries back into that source to fetch necessary data. Better choice for large Datasets because first, it doesn't take up additional space, second, it's faster.

Live Connection: much like DirectQuery mode, but connects to Analysis Services or Power BI Datasets

3. Explain deployment pipelines in Power BI Online. What stages do they include?

In Power BI, a **deployment pipeline** is a feature that helps you manage the **lifecycle of content** (datasets, reports, dashboards, and dataflows) as it moves from **development** → **test** → **production**.

Development – authors build, and edit reports/datasets

Test – Used for testing, validation, QA

Production – final version for end users

4. How can Power BI Service integrate with Microsoft Teams or SharePoint for collaboration?

To integrate Power BI Service with Microsoft Teams, install the Power BI app in Teams from the Apps menu, which provides a dedicated Power BI experience within Teams.

1. Embedding Power BI Reports in SharePoint

In SharePoint Online, you can embed Power BI reports directly into modern pages using the Power BI web part.

Users see the report inside SharePoint without opening Power BI separately.

Permissions are respected: if a user doesn't have access to the report in Power BI, they won't see it in SharePoint.

👉 Use case: A project site or department portal that displays relevant dashboards.

♦ 2. Storing and Connecting to Data in SharePoint

Power BI can connect directly to SharePoint lists or document libraries.

Connector: SharePoint Online List → import structured list data.

Connector: SharePoint Folder → read multiple Excel/CSV files stored in a library (e.g., monthly uploads).

Supports both Import and DirectQuery modes (with limitations).

👉 Use case: Store team data in SharePoint lists and visualize it in Power BI.

5. What is the XMLA endpoint in Premium and how does it benefit developers or enterprise BI teams?

The **XMLA endpoint** in Power BI is an interface that allows you to connect directly to the **Analysis Services engine** behind your Power BI datasets in the Power BI Service.

Suppose you have a huge Power BI dataset hosted in a Premium workspace.

- With XMLA endpoint, you can connect from **Excel** → PivotTable → “From Analysis Services” and point to the XMLA URL.
- Now you can analyze the dataset directly, without re-importing data.

6. Describe how usage metrics and audit logs work in Power BI Service.

Usage metrics in PBI Service are basically reports for reports that show **who is viewing reports/dashboards, how often, and when**.

What you see:

- Number of **views** (per day, week, month).
- Number of **unique viewers**.
- Which users are interacting the most.
- Which content is most popular in the workspace.

Audit logs in PBI Service are Detailed logs of **all user and admin activities** that keep track of what anyone in the organizations did inside the workspace

7. How do you manage workspace access and permissions for different users?

To manage workspace access owners can add members into that workspace by their Power BI email address. To manage permissions for different users, owners can assign a role for each member of the workspace that determines what they can do.

What you see:

- User sign-ins and activity.
- Report, dataset, dashboard creation, modification, deletion.
- Sharing, exports, downloads, and data refresh operations.
- Admin actions (e.g., changing workspace roles).

8. How can data governance be enforced in Power BI Service?

Data governance in Power BI Service is enforced through a mix of:

- **Access control** (workspaces, roles, sharing, apps)
- **Data security** (RLS, OLS, sensitivity labels)
- **Monitoring & compliance** (audit logs, admin policies)
- **Trust & control** (certified datasets, deployment pipelines)

9. What are the limitations of Row-Level Security when using DirectQuery or Live Connection?

- **DirectQuery** → You can use RLS in Power BI, but it may be slow and depends on identity passing.
- **Live Connection** → You can't define RLS in Power BI; it must come from Analysis Services.
- **Both** → RLS doesn't replace source-level security and can cause performance hits.

👉 In plain words:

RLS works, but with **DirectQuery it can be slow**, and with **Live Connection you can't set it up in Power BI at all** — it has to come from the source.

10. Explain how you can refresh a dataset via Power Automate or REST API.

Power Automate:

- In **Power Automate**, create a new **flow**.
- Choose a trigger (e.g., *When a file is updated in SharePoint*, or *Run on a schedule*).
- Add the action: **Power BI** → **Refresh a dataset**.
- Select your **Workspace** and **Dataset**.
- Save the flow → it will now trigger refreshes automatically.

REST API:

- Register an **Azure AD app** (so you can authenticate).
- Get an **access token** (OAuth2).
- Make a `POST` request to the endpoint above.
- Power BI Service queues the dataset refresh.